Manzil JEE - Practice (2025)

Binomial Theorem

Most Important JEE PYQs

Single Correct Type Questions

1. Among the statements:

(S_1) : $2023^{2022} - 1999^{2022}$ is divisible by 8.

(S_2) : $13(13)^n - 11n - 13$ is divisible by 144 for infinitely many $n \in N$. **[6 April, 2023 (Shift-II)]**

(*a*) both (S_1) and (S_2) are incorrect
(*b*) only (S_2) is correct
(*c*) both (S_1) and (S_2) are correct
(*d*) only (S_1) is correct

2. The absolute difference of the coefficients of x^{10} and x^7 in the expansion of $\left(2x^2 + \frac{1}{2x}\right)^{11}$ is equal to **[8 April, 2023 (Shift-II)]**

(*a*) $12^3 - 12$ (*b*) $11^3 - 11$
(*c*) $10^3 - 10$ (*d*) $13^3 - 13$

3. Let $x = (8\sqrt{3}+13)^{13}$ and $y = (7\sqrt{2}+9)^9$. If $[t]$ denotes the greatest integer $\leq t$, then **[30 Jan, 2023 (Shift-II)]**

(*a*) $[x] + [y]$ is even
(*b*) $[x]$ is odd but $[y]$ is even
(*c*) $[x]$ is even but $[y]$ is odd
(*d*) $[x]$ and $[y]$ are both odd

4. $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by **[8 April, 2023 (Shift-II)]**

(*a*) 34 but not by 14 (*b*) both 14 and 34
(*c*) neither 14 nor 34 (*d*) 14 but not by 34

5. If the coefficients of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax - \frac{1}{3bx^2}\right)^{11}$ are equal, then **[6 April, 2023 (Shift-II)]**

(*a*) $64ab = 243$ (*b*) $729ab = 32$
(*c*) $243ab = 64$ (*d*) $32ab = 729$

6. If the coefficients of three consecutive terms in the expansion of $(1 + x)^n$ are in the ratio 1 : 5 : 20, then the coefficient of the fourth term is **[8 April, 2023 (Shift-I)]**

(*a*) 2436 (*b*) 5481
(*c*) 1827 (*d*) 3654

7. The term independent of x in the expansion of

$$\left(1 - x^2 + 3x^3\right)\left(\frac{5}{2}x^3 - \frac{1}{5x^2}\right)^{11}, x \neq 0 \text{ is :}$$

[28 June, 2022 (Shift-II)]

(*a*) $\frac{7}{40}$
(*b*) $\frac{33}{200}$
(*c*) $\frac{39}{200}$
(*d*) $\frac{11}{50}$

8. The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is **[27 July, 2022 (Shift-I)]**

(*a*) 0 (*b*) 1
(*c*) 2 (*d*) 6

9. For the natural numbers m, n, if $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \ldots a_{m+n} y^{m+n}$ and $a_1 = a_2 = 10$, then the value of $(m+n)$ is equal to : **[20 July 2021 (Shift-II)]**

(*a*) 64 (*b*) 88
(*c*) 80 (*d*) 100

10. The value of $-{}^{15}C_1 + 2\cdot{}^{15}C_2 - 3\cdot{}^{15}C_3 + \ldots - 15.{}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \ldots + {}^{14}C_{11}$ is **[24 Feb, 2021 (Shift-I)]**

(*a*) 2^{14} (*b*) $2^{13} - 13$ (*c*) $2^{16} - 1$ (*d*) $2^{13} - 14$

11. Let $\alpha > 0$, $\beta > 0$ be such that $\alpha^3 + \beta^2 = 4$. If the maximum value of the term independent of x in the binomial expansion of $(\alpha x^{1/9} + \beta x^{-1/6})^{10}$ is $10\,k$, then k is equal to: **[2 Sep, 2020 (Shift I)]**

(*a*) 84 (*b*) 176 (*c*) 336 (*d*) 352

12. The number of ordered pairs (r, k) for which $6.{}^{35}C_r = (k^2 - 3).\,{}^{36}C_{r+1}$, where k is an integer, is: **[7 Jan, 2020 (Shift-II)]**

(*a*) 3 (*b*) 6 (*c*) 4 (*d*) 2

13. The coefficient of x^7 in the expression $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + ...+ x^{10}$ is: **[7 Jan, 2020 (Shift-II)]**

(*a*) 210 (*b*) 330 (*c*) 420 (*d*) 120

14. The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is: **[12 April, 2019 (Shift-I)]**

(*a*) – 84 (*b*) 84 (*c*) 126 (*d*) – 126

15. The value of $\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + ... + \frac{1}{49!2!} + \frac{1}{51!1!}$ is **[1 Feb, 2023 (Shift-I)]**

(*a*) $\frac{2^{50}}{50!}$ (*b*) $\frac{2^{50}}{51!}$

(*c*) $\frac{2^{51}}{51!}$ (*d*) $\frac{2^{51}}{50!}$

16. If $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + ... + 30({}^{30}C_{30})^2 = \frac{\alpha \cdot 60!}{(30!)^2}$, then α is equal to **[24 Jan, 2023 (Shift-II)]**

(*a*) 30 (*b*) 60

(*c*) 15 (*d*) 10

17. If $\frac{1}{n+1}{}^nC_n + \frac{1}{n}{}^nC_{n-1} + ... + \frac{1}{2}{}^nC_1 + {}^nC_0 = \frac{1023}{10}$ then n is equal to **[12 April, 2023 (Shift-I)]**

(*a*) 6 (*b*) 9

(*c*) 8 (*d*) 7

18. If $\sum_{k=1}^{31}({}^{31}C_k)({}^{31}C_{k-1}) - \sum_{k=1}^{30}({}^{30}C_k)({}^{30}C_{k-1}) = \frac{\alpha(60!)}{(30!)(31!)}$, where $\alpha \in R$, then the value of 16α is equal to **[28 June, 2022 (Shift-I)]**

(*a*) 1411 (*b*) 1320 (*c*) 1615 (*d*) 1855

19. If the constant term in the expansion of $\left(3x^3 - 2x^2 + \frac{5}{x^5}\right)^{10}$ is $2^k.\,\ell$, where ℓ is an odd integer, then the value of k is equal to : **[29 June, 2022 (Shift-I)]**

(*a*) 6 (*b*) 7

(*c*) 8 (*d*) 9

20. The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$ where $x \in (0, 1)$ is: **[26 Feb, 2021 (Shift-I)]**

(*a*) $\frac{10!}{\sqrt{3}(5!)^2}$ (*b*) $\frac{2.10!}{3(5!)^2}$

(*c*) $\frac{10!}{3(5!)^2}$ (*d*) $\frac{2.10!}{3\sqrt{3}(5!)^2}$

Integer Type Questions

21. The remainder when $19^{200} + 23^{200}$ is divided by 49, is ______. **[1 Feb, 2023 (Shift-I)]**

22. The number of integral terms in the expansion of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ is equal to **[11 April, 2023 (Shift-I)]**

23. If the term without x in the expansion of $\left(x^{\frac{2}{3}} + \frac{\alpha}{x^3}\right)^{22}$ is 7315, then $|\alpha|$ is equal to ______. **[1 Feb, 2023 (Shift-II)]**

24. Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}:1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to...... **[29 July, 2022 (Shift-I)]**

25. The term independent of 'x' in the expression of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}}\right)^{10}$, where $x \neq 0, 1$ is equal to. **[25 July, 2021 (Shift-I)]**

26. If the coefficient of $a^7 b^8$ in the expansion of $(a + 2b + 4ab)^{10}$ is $K.2^{16}$, then K is equal to _____. **[31 Aug, 2021 (Shift-II)]**

27. Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to ____. **[4 Sep, 2020 (Shift-I)]**

28. If $C_r = {}^{25}C_r$ and $C_0 + 5.C_1 + 9.C_2 + ... + (101).C_{25} = 2^{25}.\ k$, then k is equal to ______ **[9 Jan, 2020 (Shift-II)]**

29. If $({}^{40}C_0) + ({}^{41}C_1) + ({}^{42}C_2) + ... + ({}^{60}C_{20}) = \frac{m}{n}\,{}^{60}C_{20}$, m and n are coprime, then $m + n$ is equal to ______. **[26 June, 2022 (Shift-II)]**

30. Let $\binom{n}{k}$ denote ${}^{n}C_k$ and $\begin{bmatrix} n \\ k \end{bmatrix} = \begin{cases} \binom{n}{k}, & \text{if } 0 \le k \le n \\ 0, & \text{otherwise} \end{cases}$

If $A_k = \sum_{i=0}^{9} \binom{9}{i} \begin{bmatrix} 12 \\ 12-k+i \end{bmatrix} + \sum_{i=0}^{8} \binom{8}{i} \begin{bmatrix} 13 \\ 13-k+i \end{bmatrix}$ and $A_4 - A_3 = 190p$, then p is equal to ____________. **[26-Aug-2021(Shift-II)]**